

Auto Ticket Classification using Flow Designer

ServiceNow Project Report

A practical no-code approach for automatically classifying school IT support tickets.

Prepared as project documentation

1. Project Overview

A school IT helpdesk can receive many requests in a single day—Wi-Fi problems, projector issues, password or login difficulties, and computers that are running slowly. In the existing process, an IT staff member has to read each request and decide the correct category and subcategory manually. That adds repetitive work and can slow down ticket routing.

This project uses ServiceNow Flow Designer to make that first classification step automatic. The flow checks the wording of the ticket and assigns the appropriate Category and Subcategory when a matching keyword is found.

Project objectives

- Automatically classify a ticket when it is created.
- Reduce the amount of manual classification required from IT staff.
- Make ticket routing more consistent.
- Keep the solution no-code and easy to maintain.
- Send a confirmation email to the caller after ticket creation.

Business requirements

- Classify tickets from the issue description.
- Set both Category and Subcategory automatically.
- Use dependent choices so Subcategory options match the selected Category.
- Store ticket details in a structured format.
- Keep the design suitable for future extensions.

2. Phase 1 – Requirement Analysis & Planning

The first phase focuses on understanding the helpdesk problem and deciding what the system should do. The main requirement is simple: a student or teacher should be able to describe the issue without having to understand the internal ticket classification system.

Expected ticket categories

Issue type	Category	Subcategory
Wi-Fi / network issue	Network	Wi-Fi
Projector issue	Hardware	Projector
Password / login issue	Access	Forgot Password
Slow / hanging computer	Performance	Slow Computer

Update Set

- Create a local update set named **Project Update Set**.
- Keep the state as **In progress** while development is underway.
- Use the Global application and make the new set the current update set.

Activity : 1 Business Use Case.

Business Requirements

The system must:

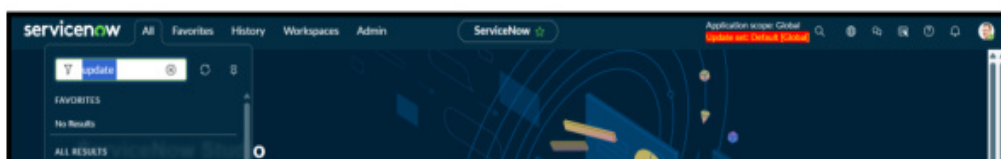
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save** the **Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.



Original implementation reference: creating and configuring the project update set.

3. Phase 2 – Backend Development & Configuration

A custom table is used to store the school IT ticket records. The table is named **Incident WorkFlow** in the supplied project material.

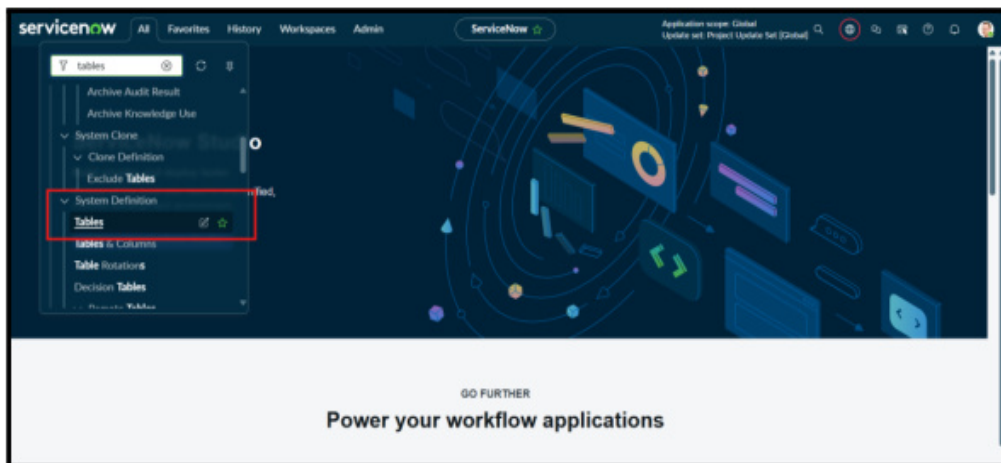
Table setup

- Open System Definition → Tables and create a new table.
- Label the table **Incident WorkFlow**.
- Leave Create module unchecked.
- Enable Auto Number from the table controls.
- Save the table and continue with field configuration.

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.
- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.

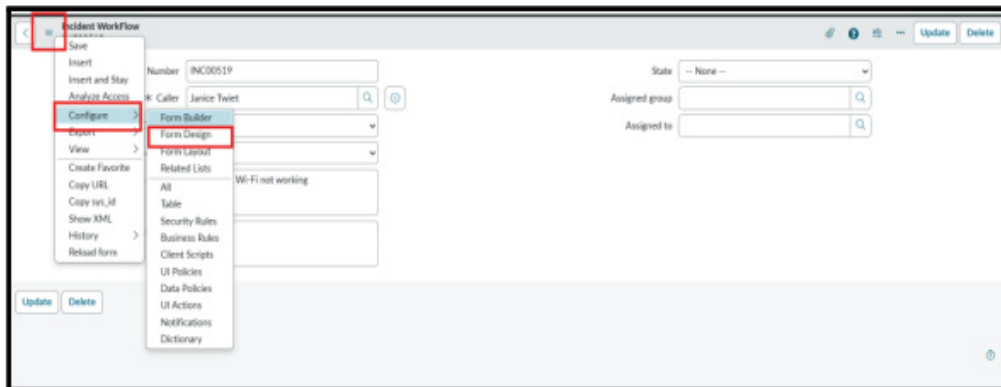


Original implementation reference: custom table creation.

4. Field Design

The table fields are designed to keep ticket information consistent. Choice fields limit the values that can be entered, while reference fields connect the ticket to users and groups.

Field	Type	Values / Reference
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	—
Description	String	—
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user



Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access,Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Compter]
Short Description	String	—
Description	String	—
State	Choice	[New,In progress,On hold,Resolved,Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

Original implementation reference: field creation and data type configuration.

5. Category–Subcategory Dependency

The Subcategory field is made dependent on Category. This keeps the form easier to use because only the relevant Subcategory choices are shown for the selected Category.

- Open the Subcategory field and choose Configure Dictionary.
- Switch to the Advanced view.
- Enable **Use dependent field**.
- Set the dependent field to **Category**.
- Map each Subcategory to its Category value.

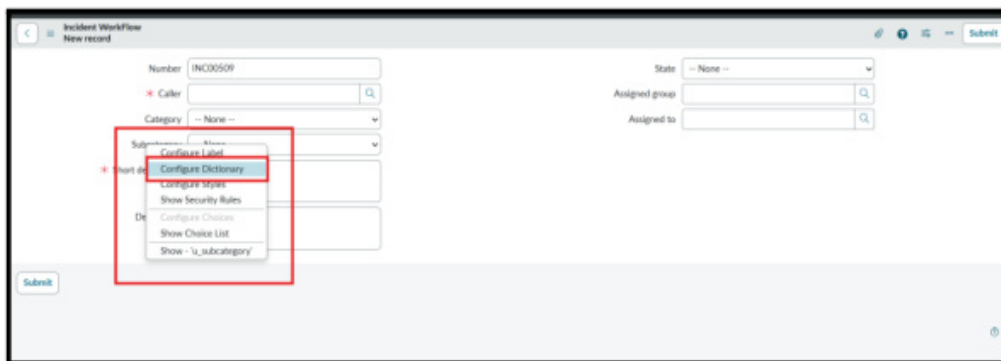
Subcategory	Depends on Category
Wi-Fi	Network
Projector	Hardware
Forgot Password	Access
Slow Computer	Performance

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

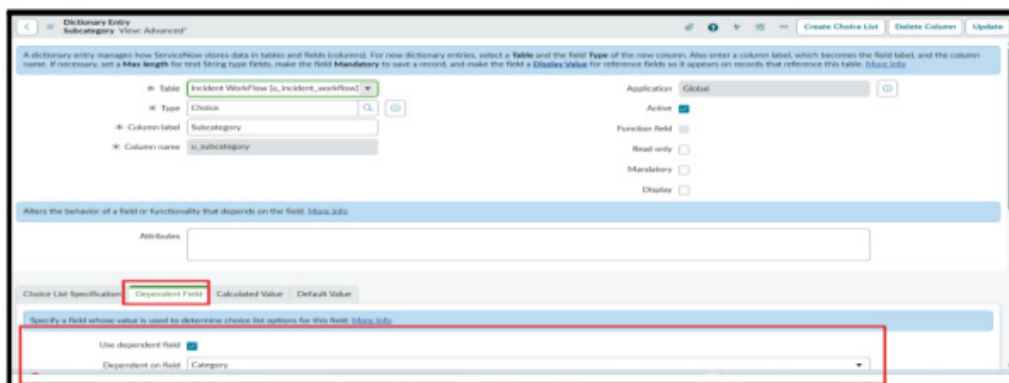
The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the **Dependent Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.



This screenshot shows the 'Incident Workflow' form in 'New record' mode. The 'Subcategory' field is highlighted with a red box, and a context menu is open over it. The menu options are: 'Configure Label', 'Configure Dictionary' (highlighted with a red box), 'Configure Styles', 'Show Security Rules', 'Configure Choices', 'Show Choice List', and 'Show - 'u_subcategory''. Other fields visible include 'Number' (INC00509), 'Caller', 'Category' (None), 'State' (None), 'Assigned group', and 'Assigned to'.



This screenshot shows the 'Dictionary Entry' configuration screen for the 'Subcategory' field. The 'Table' is set to 'Incident Workflow [u_incident_workflow]' and the 'Type' is 'Choice'. The 'Column label' is 'Subcategory' and the 'Column name' is 'u_subcategory'. The 'Advanced' tab is selected, and the 'Dependent Field' section is highlighted with a red box. In this section, the checkbox 'Use dependent field' is checked, and the 'Dependent on field' is set to 'Category'. Other tabs include 'Choice List Specification', 'Calculated Value', and 'Default Value'.

Original implementation reference: configuring the dependency between Category and Subcategory.

6. Phase 3 – Automation with Flow Designer

The main automation is built in Flow Designer. When a new ticket is created, the flow checks the Short Description for a small set of predefined keywords. If a keyword matches, the flow updates the ticket with the corresponding Category and Subcategory.

Flow setup

- Open Process Automation → Flow Designer.
- Create a new Flow named **Auto Classify School IT Tickets**.
- Use the Global application.
- Build the flow and configure the trigger as **Record Created**.
- Use the **Incident WorkFlow** table.
- Run the flow when Category is empty.

Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.

Original implementation reference: Flow Designer configuration and automation overview.

7. Classification Logic

The flow uses a simple sequence of conditions. The logic documented in the project is:

Keyword / phrase	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot Password	Access	Forgot Password
Slow Computer	Performance	Slow Computer

Let's get the details for your flow

Flow name  Auto Classify School IT Tickets

Application  Global

Description  Describe your flow.

[Show additional properties](#)

Cancel Build flow

Activity : 2 Trigger Configuration.

- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**

Workflow Studio Auto Classify School IT Tickets

Auto Classify School IT Tickets

TRIGGER

Incident Workflow Created where (Category is empty)

Trigger Created

Table Incident Workflow (u_incident_w...)

Condition All of these conditions must be met

Category is empty

New Criteria

Advanced Options

Delete Cancel Done

Data

Flow Variables

Trigger - Record Created

Incident Workflow Record

Incident Workflow Table

Run Start Time UTC

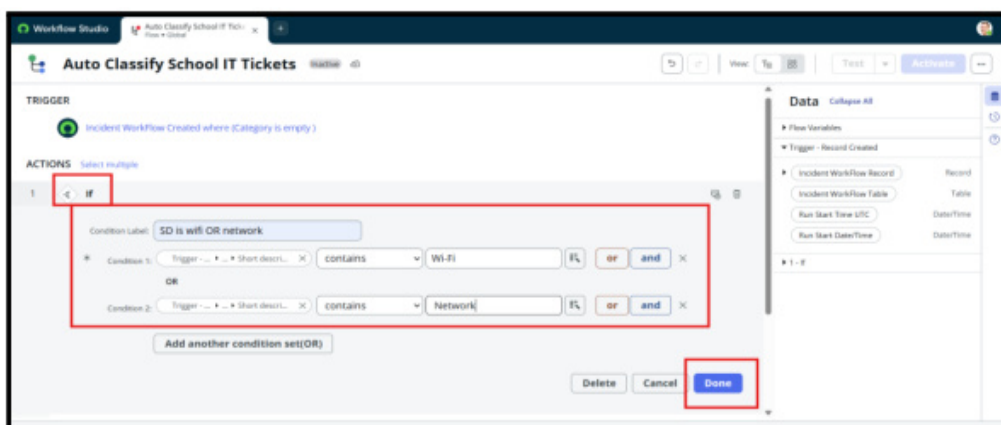
Run Start Date/Time

- Now Under **Actions** Select **Flow Logic** Option.

Original implementation reference: Wi-Fi / Network classification condition.

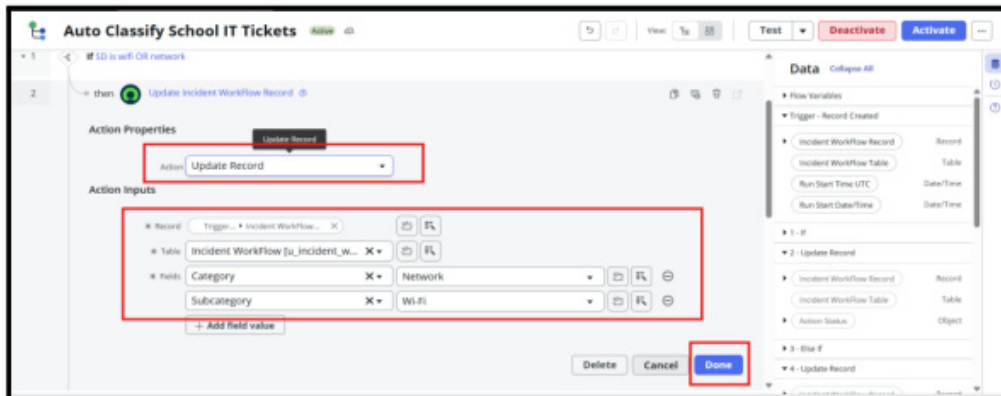
Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.



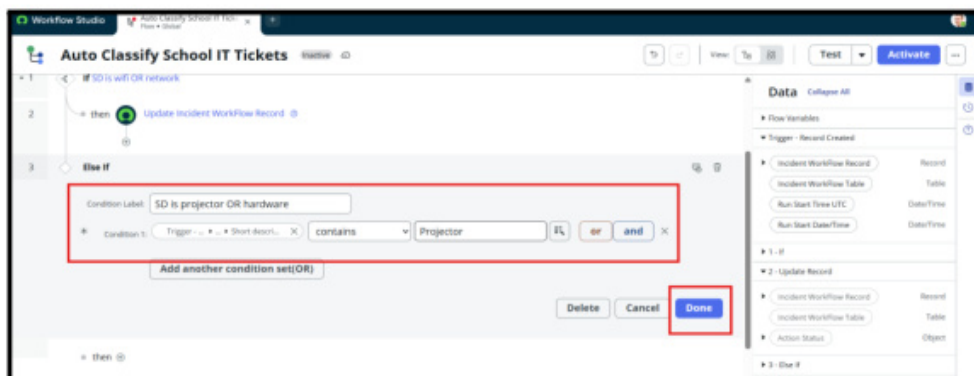
- Now Click **Action to Update record**.
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category = Network**.
- Set **Subcategory = Wi-Fi**.
- Then Click the **Done** button.

Original implementation reference: Projector / Hardware classification.



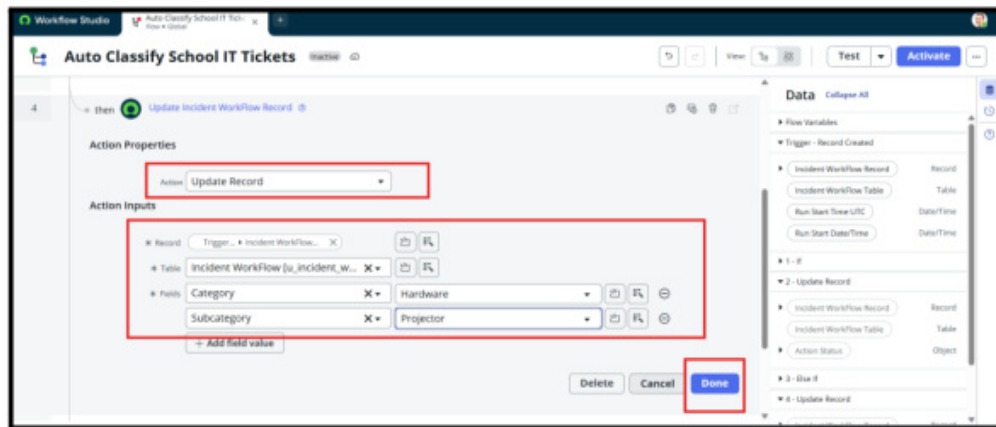
Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : **SD is Projector or Hardware.**
- Condition : **Short description Contains Prajector.**
- Then Click the **Done** button.
- Now Action is an **Update record.**
- Set **Category = Hardware.**



- Set **Subcategory = Projector.**

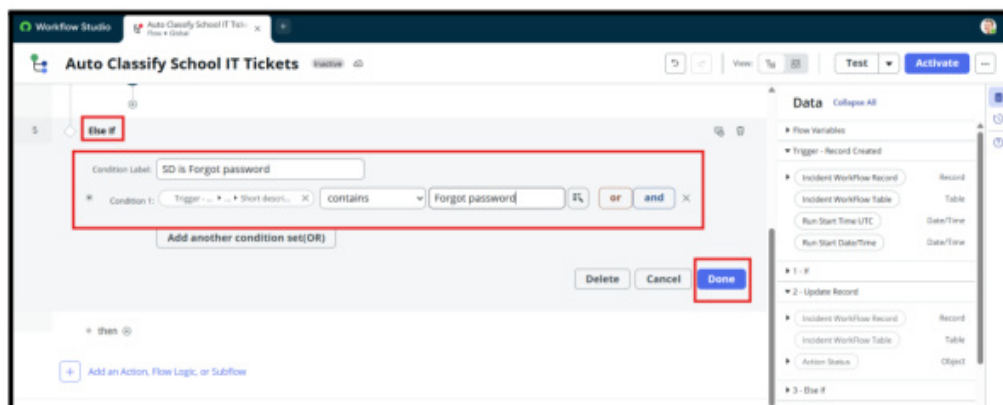
Original implementation reference: Password / Access classification.



- Then Click the **Done** button.

Activity : 5 Password Issue Classification.

- Now again select **Else If** condition.
- Condition label : **SD is Forgot password.**
- Condition : **Short description contains Forgot password.**
- Then Click the **Done** button.
- Now Action is an **Update Record.**
- Set **Category = Access.**
- Set **Subcategory = Forgot Password.**
- Then Click the **Done** button.

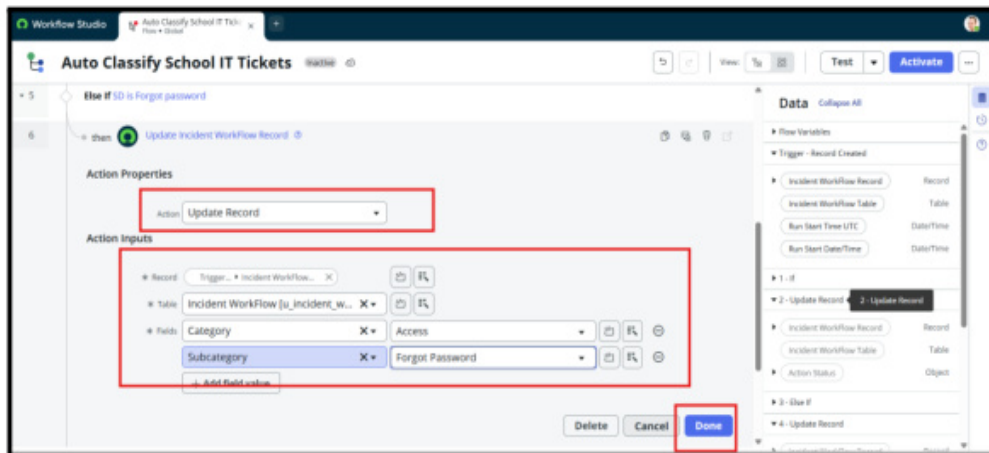


Original implementation reference: Slow Computer / Performance classification.

8. Email Notification

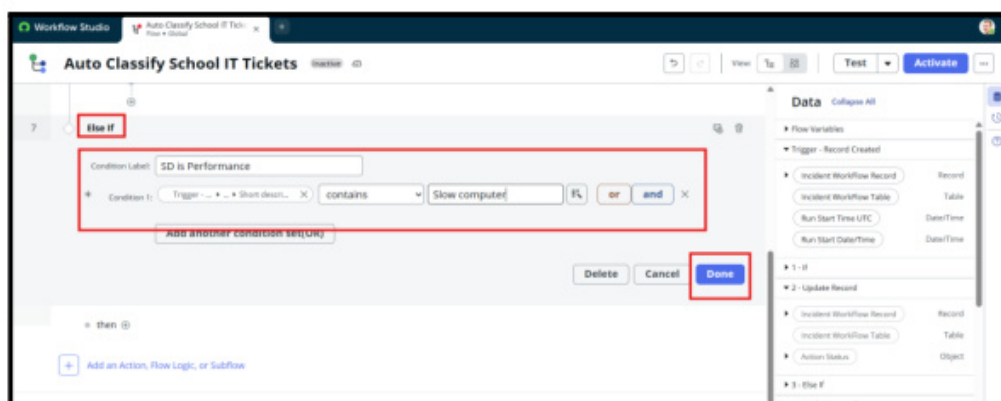
After the ticket is classified, the flow sends an email to the caller. This gives the user a simple confirmation that the request has been submitted.

- Action: **Send Email**.
- Recipient: the caller's email address.
- Subject: **Your Request for the issue has been submitted**.
- Body: a ticket confirmation message.



Activity : 6 Slow Computer Classification.

- Now Again Click the **Else If** Condition.
- Condition label : **SD is Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category = Performance**.
- Set **Subcategory = Slow computer**.
- Then Click the **Done** button.



Original implementation reference: configuring the email notification action.

9. Phase 4 – Testing & Validation

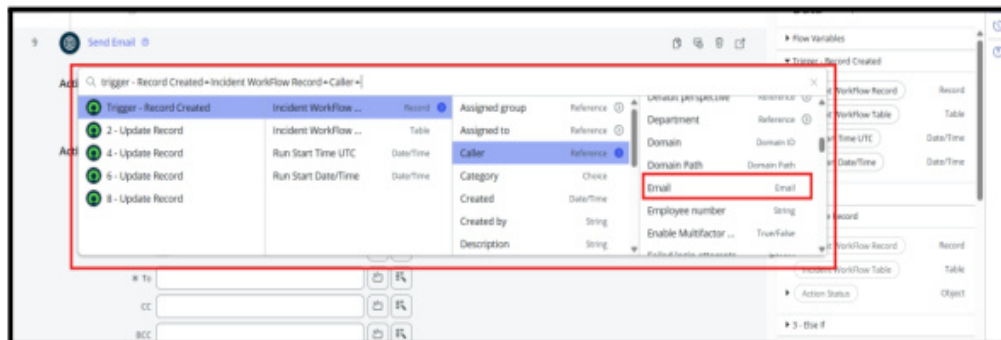
Testing checks whether the flow classifies real-looking requests correctly and whether the confirmation email is generated. The documented checks also cover data consistency and reference-field behaviour.

Validation checklist

- Mandatory fields are captured correctly.
- Auto-numbering works without duplication.
- Category and Subcategory are stored correctly.
- User and group reference fields resolve correctly.

Test scenario 1 – Wi-Fi

- Short Description: **WiFi not working in library.**
- Expected Category: **Network.**
- Expected Subcategory: **WiFi.**
- A confirmation email is sent to the caller.



- Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

Test Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library
- Then Save the Form.
- And Reload the Form.

Original implementation reference: Wi-Fi test scenario and resulting classification.

Test scenario 2 – Projector

- Short Description: **Projector not turning on.**
- Expected Category: **Hardware.**
- Expected Subcategory: **Projector.**
- A confirmation email is sent to the caller.

10. Phase 5 – Deployment & Documentation

Once development and testing are complete, the update set can be marked complete and exported. The exported XML can then be shared so the configuration can be transferred to another suitable environment.

- Open Local Update Sets and select **Project Update Set**.
- Change the state from **In progress** to **Complete**.
- Save the update set.
- Use **Export to XML** from the related links.
- The exported update-set file can be shared through the required project channel.

The screenshot shows a web form titled "Incident Workflow" with the ID "INC000912". The form contains several input fields and dropdown menus. A red box highlights the following fields: "Number" (containing "INC000912"), "Caller" (containing "Sam Savarin"), "Category" (containing "Hardware"), "Subcategory" (containing "Projector"), "Short description" (containing "Projector not turning on"), and "Description" (empty). To the right of these fields are "State" (dropdown, set to "None"), "Assigned group" (dropdown), and "Assigned to" (dropdown). At the bottom left are "Update" and "Delete" buttons. At the bottom right is a small circular icon.

PHASE 5: Deployment & Final Documentation.

- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.

Original implementation reference: deployment and update-set completion steps.

11. End-to-End Process

The complete workflow can be understood as a short sequence from user input to confirmation:

2	Flow is triggered	The record is created and Category is still empty.
3	Keywords are checked	The Short Description is evaluated against the configured conditions.
4	Ticket is classified	Category and Subcategory are updated automatically.
5	Email is sent	The caller receives a submission confirmation.

Why the approach is useful

- It removes a repetitive classification step from the agent's workflow.
- It gives the same classification result for the same configured keywords.
- Dependent choices reduce unrelated Subcategory options.
- Flow Designer keeps the implementation no-code and easier to maintain.
- The design can be extended later with additional automation.

12. Conclusion

The Auto Ticket Classification using Flow Designer project provides an end-to-end way to organize school IT support requests in ServiceNow. Instead of asking IT staff to classify every new ticket manually, the flow uses the ticket's Short Description to identify a matching issue type and update the Category and Subcategory automatically.

The project combines a structured custom table, dependent choice fields, Flow Designer conditions, record updates, and email notification. Testing with Wi-Fi and projector examples demonstrates the intended classification behaviour and confirmation process.

Because the solution is built without complex scripting or machine learning, it is straightforward to understand and maintain. The same foundation could later be extended with features such as assignment automation, SLA tracking, or predictive capabilities, depending on future project requirements.

Project status

Core ticket classification, dependency configuration, notification, testing, and update-set documentation are covered in this report.